Kursat Kara, Ph.D.

Associate Professor | AIAA Associate Fellow
Director, Kara Aerodynamics Research Laboratory

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AT A GLANCE

$3.5M+
external funding as PI or Co-PI (NSF, FAA, NASA, U.S. Navy)

$1.26M
in active awards (FAA, NASA, U.S. Navy)

84
peer-reviewed papers: 42 journal, 42 conference

14
journal papers published in 2025–2026

12
graduate students supervised (6 Ph.D., 6 M.S.), including 3 Ph.D. and 1 M.S. in progress

957
students taught in MAE 3293 since 2019; 3 new graduate courses

Funding totals: awarded external funds (OSU share), excluding computing allocations, in-kind support, and internal grants. Counts as of Oct. 2026.

Research: High-fidelity CFD (DNS and LES), physics-guided reduced-order modeling, and scientific machine learning for unsteady aerospace flows: dynamic stability of reentry capsules, ship airwakes, urban atmospheric turbulence for advanced air mobility, hypersonic boundary-layer transition, flow control, and quantum algorithms for CFD.

APPOINTMENTS AND EDUCATION

2025–present	**Associate Professor**, School of Mechanical & Aerospace Engineering, Oklahoma State University, Stillwater, OK
2019–2025	**Assistant Professor**, School of Mechanical & Aerospace Engineering, Oklahoma State University, Stillwater, OK
2010–2018	**Assistant Professor**, Aerospace Engineering Department, Khalifa University, Abu Dhabi, UAE. Founding Faculty. ABET-accredited. The first Aerospace Engineering Program in the UAE.
Education	**Ph.D.**, Aerospace Engineering, Old Dominion University, 2008; **M.Sc.** (2003) and **B.Sc.** (1999), Aeronautical Engineering, Istanbul Technical University

HONORS

2026	Associate Fellow, American Institute of Aeronautics and Astronautics (AIAA), Class of 2027 (elected August 2026)
2026	AIAA Section-Student Branch Partnership Award, Second Place, Oklahoma Section — Technical Symposium Advisor
2025	Excellent Faculty Award, College of Engineering, Architecture and Technology
2015	President's Faculty Excellence Award for Outstanding Teaching, Khalifa University

SELECTED RESEARCH FUNDING

2026–2028	**Co-PI**, FAA, ASSURE, *A94 – Effect of Turbulence on Aircraft Structural Loading of AAM VTOL Vehicles*, $772,698, Contract No. 25-C-UAS-OKSU-01, OSU share of $2.1M total award.
2026–2028	**PI**, U.S. Navy (NAWCAD), *Artificial Intelligence for Real-time Wake Inference in Ship Environments (AIRWISE)*, $199,999, Award No. G10007133, Contract No. N00421-26-1-0004
2026–2027	**PI**, NASA, Early Stage Innovations (ESI22) - Paid Extension, *Physics-Guided Multifidelity Learning for Characterization of Blunt-Body Dynamic Stability*, $291,955, Award No. 80NSSC23K0231-P00003
2023–2026	**PI**, NASA, Early Stage Innovations (ESI22), *Physics-Guided Multifidelity Learning for Characterization of Blunt-Body Dynamic Stability*, $649,879, Award No. 80NSSC23K0231
2021–2022	**Co-PI**, OCAST, AR20-020, *Next Generation Smart Heatsinks*, $45,000
2021	**PI**, NSF, REU Supplement to Award 1925147, *Safe Wind-Aware Navigation for Collaborative Autonomous Aircraft in Low Altitude Airspace REU Supplement*, $15,994, Award No. 1925147, Supported two undergraduate researchers.
2020–2025	**PI**, NSF, NRI: INT, *Safe Wind-Aware Navigation for Collaborative Autonomous Aircraft in Low Altitude Airspace*, $1,490,043, Award No. 1925147

SELECTED PUBLICATIONS

84 peer-reviewed papers (42 journal, 42 conference) and 73 conference abstracts and posters; full list on Google Scholar. **Bold**: K. Kara; underlined: advised student.

1. Kassem, A., S. A.-S. Romeo, B. Tiwari, L. Musharrat, O. San, and **K. Kara**, (Apr. 27, 2026), "Discovery of governing equations for reduced-order aerodynamic modes of atmospheric entry capsules," *Aerospace Science and Technology*, 178, Art. 112454, DOI: 10.1016/j.ast.2026.112454.
2. Tiwari, B., L. Musharrat, S. A.-S. Romeo, A. Kassem, **K. Kara**, and O. San, (Feb. 23, 2026), "Adaptive Bayesian Learning for Stability Characterization of Re-entry Vehicles," *Scientific Reports*, 16, Art. 10267, DOI: 10.1038/s41598-026-40068-7.
3. Romeo, S. A.-S., M. Alam, and **K. Kara**, (Feb. 17, 2026), "Reduced-Order Modeling of Sweeping Jet Actuators Using Eigenvalue-Sorted Dynamic Mode Decomposition," *Aerospace*, 13(2), Art. 194, DOI: 10.3390/aerospace13020194.

4. Kassem, A., S. A.-S. Romeo, B. Tiwari, O. San, and **K. Kara**, (Jan. 3, 2026), "Data-Driven Reduced-Order Modeling Framework for Blunt-Body Aerodynamic Characterization," *Aerospace Science and Technology*, 171, Art. 111615, DOI: 10.1016/j.ast.2025.111615.
5. Oz, F., S. A.-S. Romeo, A. Kassem, D. Ekelschot, J. Schulz, C. Kazemba, O. San, and **K. Kara**, (Oct. 21, 2025), "Nonlinear Parameter Estimation for Entry Capsule Dynamic Stability Analysis and Uncertainty Quantification," *Journal of Spacecraft and Rockets*, 63(2), pp. 407–420, DOI: 10.2514/1.A35998.
6. Romeo, S. A.-S., A. Kassem, B. Tiwari, O. San, and **K. Kara**, (Oct. 11, 2025), "Multifidelity Data Fusion for Analyzing Dynamic Stability of Atmospheric Entry Vehicles," *Aerospace Science and Technology*, 168, Art. 111049, DOI: 10.1016/j.ast.2025.111049.
7. Krawczyk, Z., R. K. S. S. Vuppala, R. Paul, and **K. Kara**, (May 10, 2025), "**[FEATURE PAPER]** Urban Wind Field Effects on the Flight Dynamics of Fixed-Wing Drones," *Drones*, 9(5), Art. 362, DOI: 10.3390/drones9050362.
8. Oz, F. and **K. Kara**, (Jul. 10, 2024), "Controlling Hypersonic Boundary Layer Transition with Localized Cooling and Metasurface Treatments," *Scientific Reports*, 14, Art. 15928, DOI: 10.1038/s41598-024-66867-4.
9. Vuppala, R. K. S. S., Z. Krawczyk, R. Paul, and **K. Kara**, (Jan. 3, 2024), "Modeling Advanced Air Mobility Aircraft in Data-Driven Reduced-Order Realistic Urban Winds," *Scientific Reports*, 14, Art. 383, DOI: 10.1038/s41598-023-50719-8.
10. Oz, F., O. San, and **K. Kara**, (May 12, 2023), "An Efficient Quantum Partial Differential Equation Solver with Chebyshev Points," *Scientific Reports*, 13, Art. 7767, DOI: 10.1038/s41598-023-34966-3.

TEACHING

- **MAE 3293** Fundamentals of Aerodynamics; undergraduate; every fall, 2019–2026; 957 students (91–162 per offering)
- **MAE 5010** special topics; graduate; new courses developed: Quantum Computational Fluid Dynamics (Spring 2021); Hydrodynamic Stability and Transition to Turbulence (Spring 2022)
- **MAE 5020** Data-Driven Methods in Aerospace Engineering; graduate; new course developed; every spring, 2024–2026; 52 students (16–20 per offering)
- **MAE 5943** Unsteady Aerodynamics and Aeroacoustics; graduate; course materials updated; Fall 2020, Spring 2023; 22 students (10–12 per offering)
- **Khalifa University, 2010–2018:** 14 undergraduate and graduate courses, including Introduction to CFD, Boundary Layer Analysis, Numerical Methods in Aero/Fluids, Low-Speed Aerodynamics with Laboratory, and High-Speed Aerodynamics.
- **Curriculum funding:** ANSYS, *Classical Aerodynamics Meet with Modern Engineering Simulation Tools* (2024).

STUDENT MENTORING AND OUTCOMES

- **Ph.D. graduates:** Rohit Vuppala (Ph.D. 2024), Senior Researcher of Applied ML, Fujitsu Research; Furkan Oz (Ph.D. 2024), R&D Associate, Oak Ridge National Laboratory; Bartosz Jurewicz Slupski (Ph.D. 2019), Sr. R&D Programs & Engagement Manager at ASPIRE.
- **M.S. graduates:** Mobashera Alam (M.S. 2022), Flow Assurance Manager, JACAM Catalyst, Gardendale, TX; Abdelnasir Alnaqbi (M.S. 2020), Secondee at Kearney; Shaikha Mohammed Al-Ali (M.S. 2019), Systems Integration Engineer, Ministry of Defence, UAE; Shamma Zaal Al Hemeiri (M.S. 2019), Aerospace Engineer, Atlas Aerospace, UAE; Amani M. Bin Amro (M.S. 2019), Principal Engineer, EDGE, UAE.
- **Current:** Ashraf Kassem (Ph.D., since 2023); Shafi Al-Salman Romeo (Ph.D., since 2023); Mashrur Shejan (Ph.D., since 2025); Sabbir Hossain (M.S., since 2025); 29 undergraduate researchers mentored at OSU since 2019.
- **Undergraduate placements:** Peyton Pierson, SpaceX, Build Engineer working on Starship; Tyler R. Landua, Boeing.
- **Student honors:** Ashraf Kassem: Dean's Overall Outstanding Graduate Student Award (2026), MAE Outstanding Graduate Student Award (2026), MAE Outstanding Graduate Research Assistant Award (2025), NASA Ames Research Center Internship (2024); Shafi Al-Salman Romeo: MAE Excellent GTA Award (2025), NASA Ames Research Center Internship (2024); Rohit Vuppala: Roy and Virginia Dorrough Distinguished Graduate Fellowship (2023), Los Alamos National Laboratory Internship (2023); Furkan Oz: Best Oral Presentation (2023), NASA Ames Research Center Internship (2023).

SELECTED SERVICE

- **Editorial:** Editorial Board Member, Nature Scientific Reports (2023–present; 11 manuscripts); Guest Editor, Quantum Computing Collection, Nature Scientific Reports (2024–2025; 7 manuscripts).
- **Federal review:** NSF ACCESS Allocation Review Committee (2022–present); NSF NAIRR Pilot Review Committee (2024–present); NSF P260416 FY26 Fluids ERI Panel (2025); DOE Office of Advanced Scientific Computing Research (2022); NASA STMD TechLeap Prize: Space Technology Payload Challenge (2025); NASA EPSCoR Rapid Response Research (R3) (2025); NASA Next Gen STEM TEAM II Innovator (2024); Reviewer, Postdoctoral Fellowship – Quantum Computing, Naval Research Laboratory (2024).
- **Conference leadership:** Technical Chair, AIAA SciTech 2018 ASM (Applied Aerodynamics); Technical Co-Chair, AIAA Aviation 2020 Applied Aerodynamics Conference; Technical Chair, AIAA Aviation 2016 Applied Aerodynamics Conference; Technical Co-Chair, AIAA Aviation 2014 Applied Aerodynamics Conference.
- **Professional:** Executive Committee Member (NUGEX), NERSC Users Group (2022–2025).
- **University:** Member, CEAT Academic Council — MAE Representative (2026–present); Chair, Mechanical Engineering Faculty Search Committee (2023–2024); Member, MAE Aero Curriculum Committee (2019–present).
- **Outreach:** Chair of Organizing Committee, IBM Qiskit Fall Fest (2024, 2025).